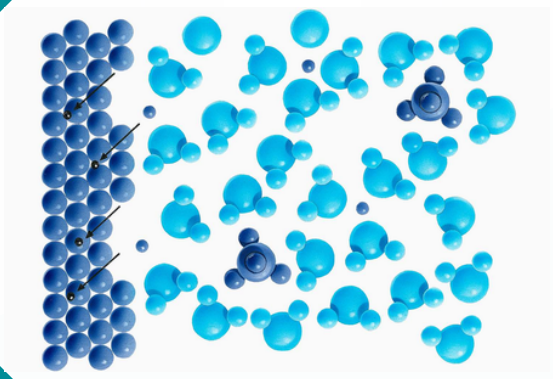
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

JEE (MAIN+ADVANCED)

Organic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Oxidation

English Medium

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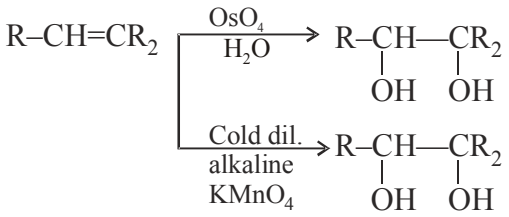
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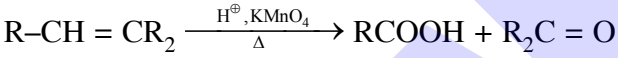
OXIDATION

(I) OXIDATION OF ALKENES



* Cold dil. alkaline $KMnO_4$ is called as Bayer's reagent, use as a test of unsaturation.
* Overall syn addition
* Given by alkenes & alkynes
* Benzene & Cyclopropane can not give this reaction.

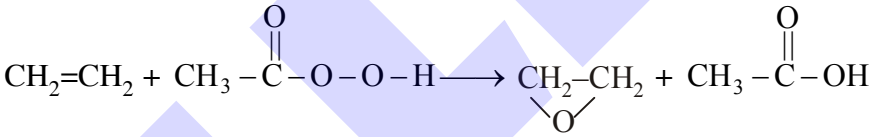
If we use acidic $KMnO_4$ or warm $KMnO_4$ or concentrated $KMnO_4$ the oxidative cleavage of Glycol occurs resulting in mixture of Carboxylic acids & Ketones.



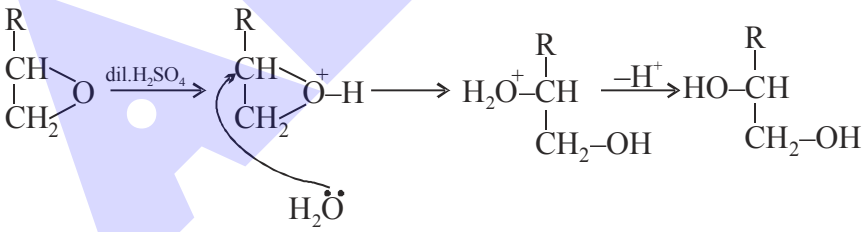
Hot acidic $KMnO_4$, Hot acidic $K_2Cr_2O_7$ & hot acidic $NaIO_4$ gives same result with alkene. The effect is similar to that of oxidative ozonolysis on alkenes.

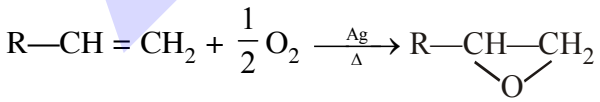
Preilschaive reaction :

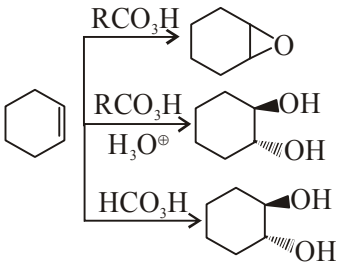
Epoxidation of alkenes is reaction of alkenes with peroxyacids.

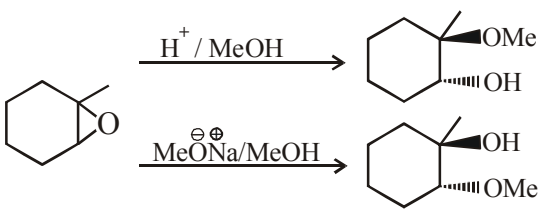


→ Rate α nucleophicity of alkene.
→ With the increase in e^- withdrawing substituents in leaving group, rate increases.
→ Hydrolysis of epoxides form anti diols.



→ 





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OXO PROCESS :

Hydroformylation/rolyen synthesis :

$$CH_2 = CH_2 \xrightarrow[\text{Lewis acid high T / P}]{CO+H_2} CH_3–CH_2–CHO$$

Carbonylation / Koch's reaction :

$$CH_2 = CH_2 \xrightarrow[\text{Lewis acid high T / P}]{CO+H_2O} CH_3–CH_2–COOH$$

$$CH_2 = CH_2 \xrightarrow[\text{Lewis acid high T / P}]{CO+ROH} CH_3–CH_2–\overset{O}{\overset{\|}{C}}–OR$$

$$HC \equiv CH \xrightarrow[\text{Lewis acid high T / P}]{CO+H_2O} \underset{\text{Acrylic acid}}{CH_2=CH–COOH}$$

$$HC \equiv CH \xrightarrow[\text{Lewis acid high T / P}]{CO+MeOH} \underset{\text{Methyl acrylate}}{CH_2=CH–\overset{O}{\overset{\|}{C}}–OMe} \xrightarrow{\text{Polymerisation}} \underset{\text{PMA}}{\left(CH_2–\underset{COOMe}{\underset{|}{CH}}\right)_n}$$

(II) OXIDATION OF ALCOHOLS

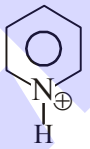
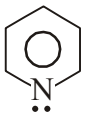
Oxidising agents

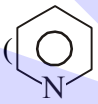
(1) $H^{\oplus}/K_2Cr_2O_7, \Delta$ (Strong oxidising agent)

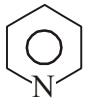
(2) $H^{\oplus}/KMnO_4, \Delta$ (Strong oxidising agent)

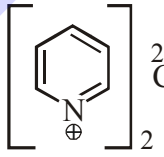
(3) **Jones reagent :** CrO_3 / H_2SO_4 treated with alcohol usually taken in acetone (strong oxidising agent)

(4) Cu / 300°C (or Red hot Cu tube)

(5) PCC (Pyridinium chloro chromate)  or 

(6) Collin's reagent ( (2 mol) + CrO_3 + CH_2Cl_2)

(7) Sarett reagent (i.e. PCC in CH_2Cl_2)  + CrO_3 + HCl + CH_2Cl_2

(8) PDC (Pyridinium dichromate) 

(9) TsCl + DMSO + $NaHCO_3$

$$RCH_2OH \xrightarrow{Ts–Cl} RCH_2OTs \xrightarrow[NaHCO_3]{DMSO} RCHO$$

$$R_2CHOH \xrightarrow{Ts–Cl} R_2CH–OTs \xrightarrow[NaHCO_3]{DMSO} R_2CO$$

$$R_3COH \xrightarrow{Ts–Cl} R_3C–OTs \xrightarrow[NaHCO_3]{DMSO} \times$$

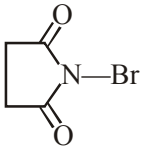
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(10) MnO_2-Oxidises only allylic or benzylic– OH. 1° Allylic or benzylic OH $\xrightarrow{MnO_2}$ Aldehyde

2° Allylic or benzylic OH $\xrightarrow{MnO_2}$ Ketone

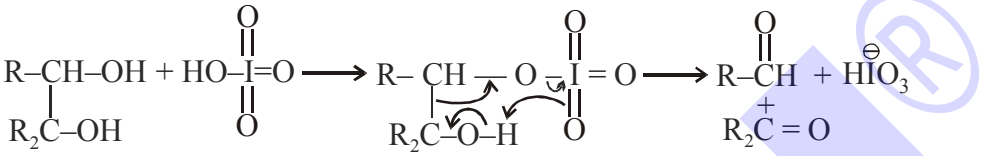
No effect on 3° ROH and on Carbon-Carbon multiple bond.

(11) NBS (N-Bromosuccinamide)



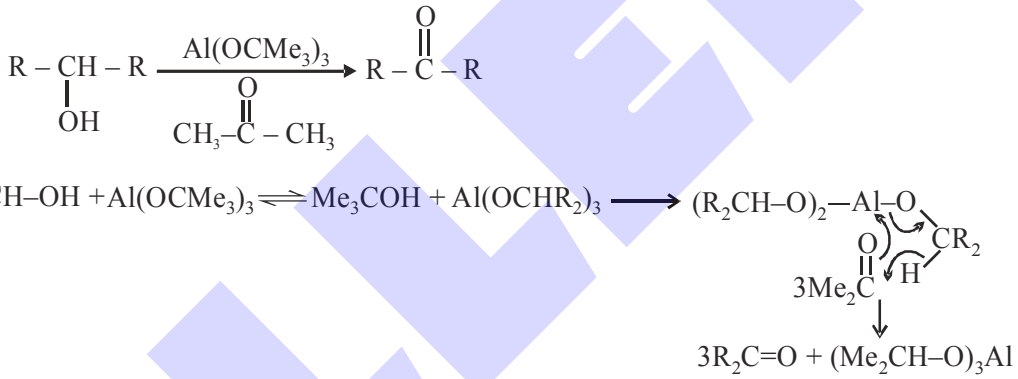
(12) **Periodic cleavage**

A similar oxidation is obtained in case of HIO_4 known as periodic cleavage.



Reaction is observed for Vic-diols.

(13) **Openaur oxidation**



Oxidation of alcohol with aluminium tertiary butoxide is Openaur oxidation.

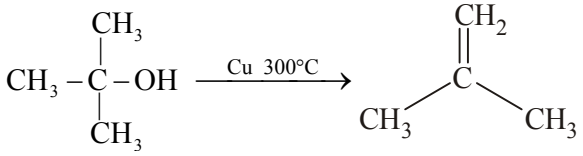
Ex. Different oxidising agents are used to oxidise alcohols in corresponding carbonyl compounds and carboxylic acids.

e.g. (I) $R-\underset{\text{1°alcohol}}{CH_2-OH} \xrightarrow{1,2,3,} R-\overset{O}{\overset{\|}{C}}-OH$

(II) $R-\underset{\text{1°alcohol}}{CH_2-OH} \xrightarrow{4,5,6,7,8,9,10,11} R-\overset{O}{\overset{\|}{C}}-H$ (Aldehyde)

(III) $R-\underset{\text{2° Alcohol}}{\overset{OH}{\overset{|}{CH}}}-R' \xrightarrow{1,2,3,4,5,6,7,8,8,9,10,11} R-\overset{O}{\overset{\|}{C}}-R'$ (Ketone)

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(IV)  Dehydration takes place.

(V) Double bond or Tripple bond is not affected by 4,5,6,7,8,9,10

(VI) No effect on 3° alcohol by 1,2,3,5,6,7,8,9,10,12,13

(III) OXIDATION OF CARBONYL COMPOUNDS

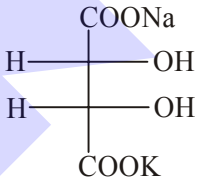
1. Tollen's Test (Silver mirror test)

$RCHO + [Ag(NH_3)_2]OH$ (Tollen's Reagent) $\xrightarrow{\Delta}$ $RCOOH + NH_3 + Ag\downarrow$ (Silver mirror)

Aldehyde acts as reducing agent, they can reduce mild oxidizing agents like Tollen's Reagent.

Tollen's test gentle Heating for 20 to 25 mins.

2. **Fehling's Solutions**



Fehling's A	Fehling's B
aq. $CuSO_4$	Alkaline solution of Roschelye salt (sodium potassium tartrate)

It act's a carrier for Cu^{2+} as it make reversible complex with Cu^{2+}

This test is also used is Blood and Urine test.

$$RCHO + Cu^{2+} \xrightarrow[\Delta]{H_2O} \underset{\downarrow}{RCOOH} + \underset{\downarrow}{Cu^{\oplus}}$$
$$\underset{\downarrow}{RCOO^-} \quad \underset{\downarrow}{Cu_2O\,(\text{red ppt.})}$$

Note : Aromatic aldehyde shows negative test with fehling reagent.

3. **Benedict's solution**

Sodium Citrate + NaOH + $NaHCO_3$ + $CuSO_4$

$$RCHO + Cu^{2+} \xrightarrow[\Delta]{H_2O} \underset{\downarrow}{RCOOH} + \underset{\downarrow}{Cu^{\oplus}}$$
$$\underset{\downarrow}{RCOO^-} \quad \underset{\downarrow}{Cu_2O\,(\text{red ppt.})}$$

It is similar to Fehling test

4. $RCHO + HgCl_2 + H_2O \xrightarrow[\Delta]{} RCOOH + 2HCl + Hg_2Cl_2$ (white)

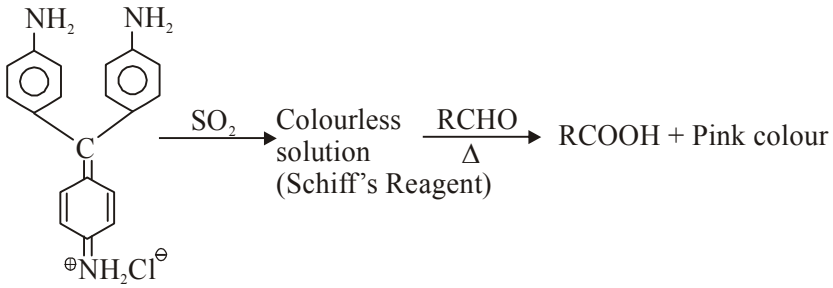
$RCHO + Hg_2Cl_2 + H_2O \xrightarrow[\Delta]{} RCOOH + 2HCl + 2Hg\downarrow$ greyish black

5. **Schiff's Reagent**

Schiff's Reagent is aq. solution of following base decolourised by passing SO_2.

Aldehyde restore pink colour of Schiff's reagent.

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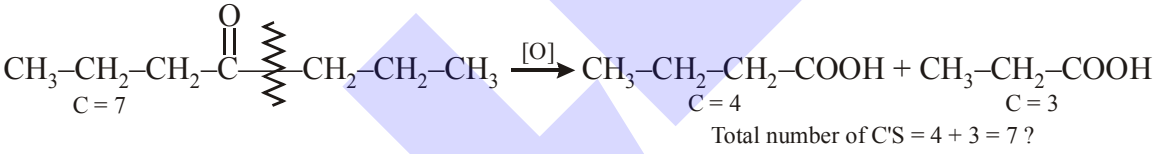
p-Rosaniline Hydrochloride
Magenta colour (Fuschin)

Ketons are not easy to oxidize so they do not give these 5 tests. These five tests can be used to distinguish aldehyde and ketones. Both gives 2,4 DNP test

Oxidation of ketones : Ketones undergo oxidation only in drastic conditions.

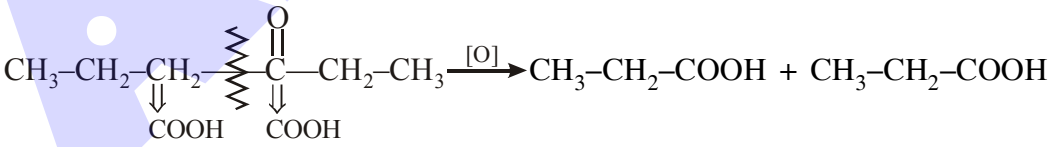
During the oxidation of ketones there is breaking of carbon-carbon bond between α-carbon and carbonyl carbon. In this process both carbons convert into carboxylic groups. This leads to the formation of two moles of monocarboxylic acids.

Case - I : Oxidation of Symmetrical ketones :

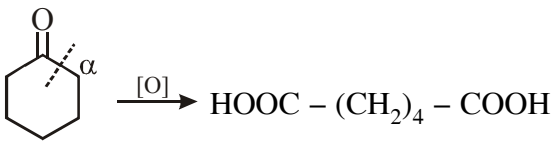


Thus number of carbons in any product is less than the number of carbons in ketone.

Case - II : Oxidation of unsymmetrical ketones : In case of unsymmetrical ketones, those —C(=O)—R bond break in which alkyl group has more number of carbons. This rule is known as **Poff's rule.**

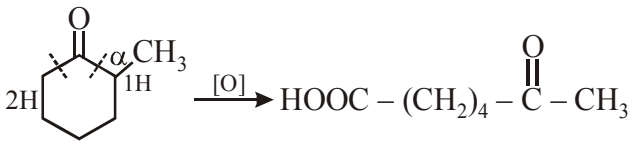


Case - III : Oxidation of cyclic ketones : Formation of dibasic acid takes place from cyclic ketones. In this case number of carbons in ketone and dibasic carboxylic acid is always same.



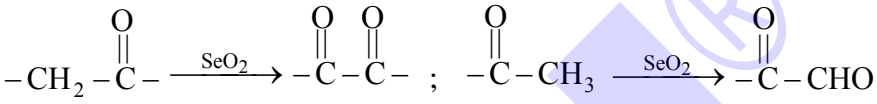
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Note : If both α-carbons are not identical then ;

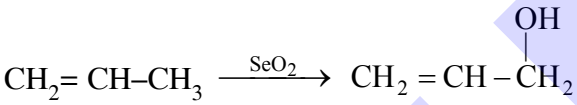


Allylic oxidation

SeO_2 is a selective oxidizing agent with converts $-CH_2-$ group adjacent to carbonyl group into carbonyl group.The reagent, in general, oxidises active methylene and methyl groups to ketonic and aldehydic groups respectively.

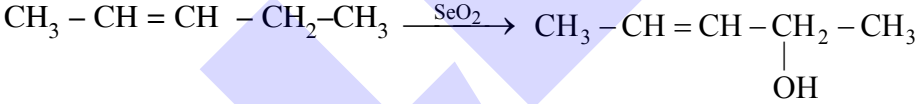


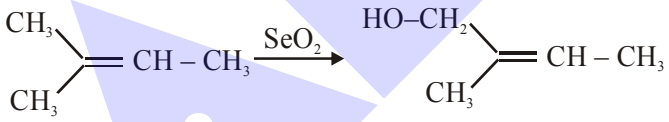
Double bonds, triple bonds and aromatic rings may also activate the methylene group.The methylene or methyl group α to the most highly substitued end of the double bond is hydroxylated according to the order of preference of oxidation $CH_2 > CH_3 > CH$ groups.

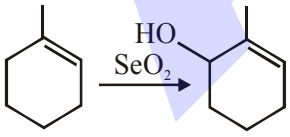


$2° C-H > 1° C-H > 3° C-H$

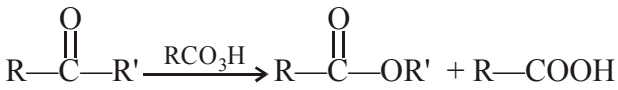
Rate of reactivity order







Baeyer villiger oxidation



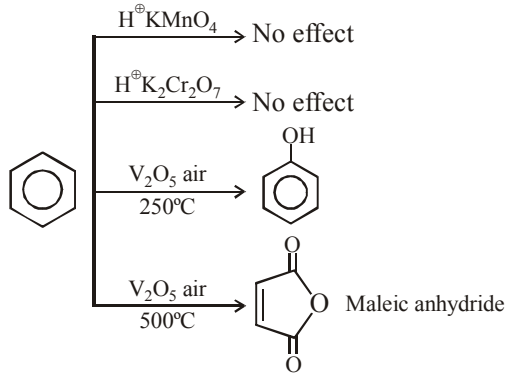
Migratory order for above reaction

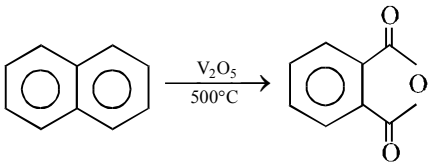
$-H > 3° > 2° > -Ph > 1° > -Me$

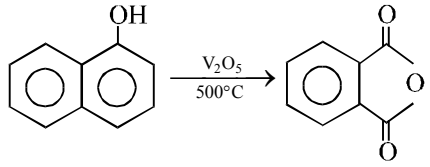
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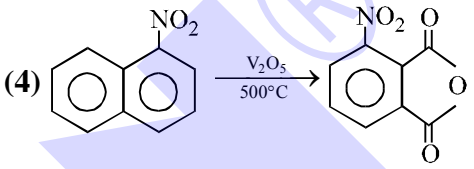
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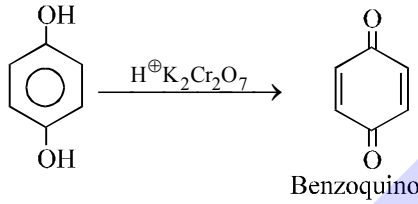
Oxidation of Aromatic Compounds.

(1) 

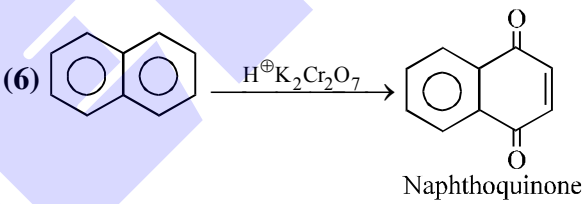
(2) 

(3) 

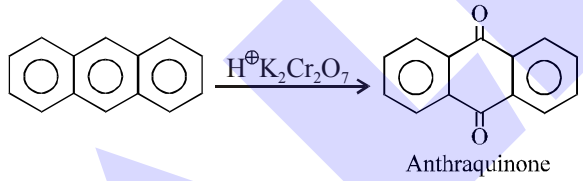
(4) 

(5) 

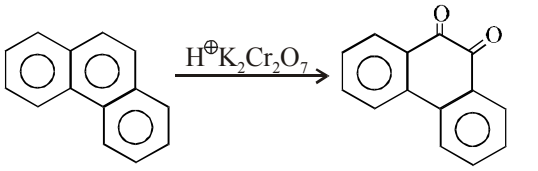
Benzoquinone

(6) 

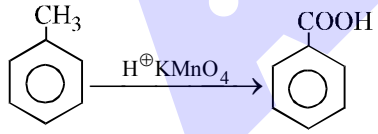
Naphthoquinone

(7) 

Anthraquinone

(8) 

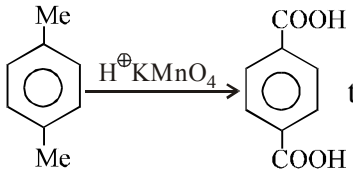
Phenanthroquinone

(9) 

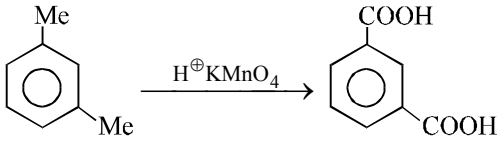
(10) $Ph\text{-}(CH_2)_nCH_3$, $PhCHMe_2$, $Ph–CH_2OH$, $Ph–CH_2Br$, $PhCHCl_2$, PhCHO, Ph-  & all the compounds having at least one α H give PhCOOH.

(11) $Ph–CMe_3 \xrightarrow[\Delta]{H^{\oplus}KMnO_4} Me_3C–COOH$

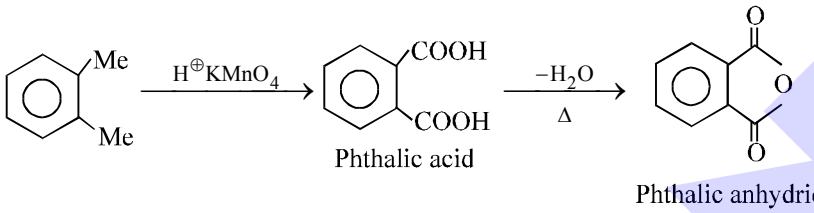
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald&Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex_p65"

(12) 

terephthalic acid

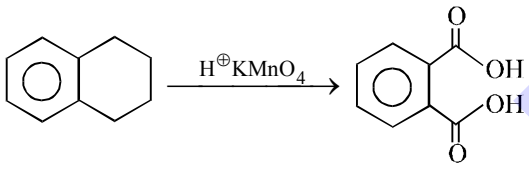
(13) 

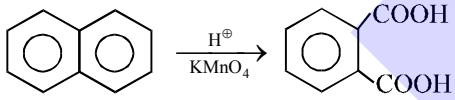
isophthalic acid

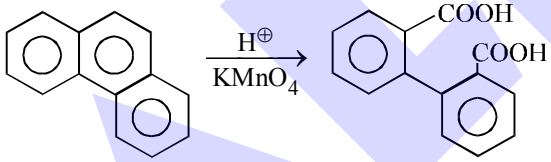
(14) 

Phthalic acid

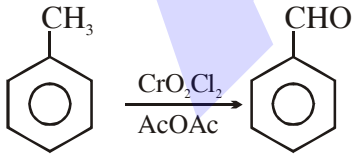
Phthalic anhydride

(15) 

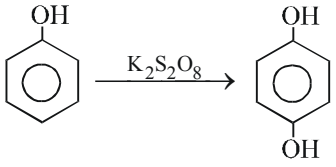
(16) 

(17) 

Etard oxidation

(18) 

Elb's persulphate oxidation:

(19) 

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdb-Alc & Et- Hydr -Carb -Oxi- Red- Ald& Ket-(OC)\Eng\05_Oxidation\01_Theory+Ex.p65"